

# DiSCoQR: Diffusion-Driven Semantic Compression for Robust Image Steganography in Standard QR Codes

Lijing Ren

rlj05@gdmec.edu.cn

Guangdong Mechanical & Electrical Polytechnic  
Guangzhou, Guangdong, China

Denghui Zhang\*

denghui.zhang@gzhu.edu.cn

Guangzhou University  
Guangzhou, Guangdong, China

## Abstract

Secure and undetectable transmission of visual intelligence through open networks presents a critical technological challenge in today's heavily monitored digital world. Text-based encryption is insufficient for massive, multimodal data from ubiquitous devices. The fundamental dilemma lies in transmitting large-scale visual data through noisy, adversarial channels while ensuring seamless concealment and robust recovery. In this paper, we bridge the gap by combining diffusion models with the universal presence and robustness of Quick Response (QR) codes for covert communication. Grounding steganographic security in the normative patterns of machine-readable symbols decouples it from the statistical properties of natural images. The reconstructions reside on the same semantic manifold as the originals, ensuring their functional equivalence in downstream tasks and maintaining robustness against quality degradation that would cripple pixel-based methods. Our training-free model compresses complex visual information into binary representations that integrate seamlessly into the data region of standard-compliant QR codes. This approach facilitates covert data transmission while preserving the full functionality and readability of the QR container. We achieve a compression ratio exceeding 400:1 while retaining semantic information for high-quality reconstruction. The preservation of all functional QR code components ensures complete compatibility with standard readers while concealing substantial visual content. Experimental results show a superior balance of security and performance compared to existing methods, all of which require no extra training.

## CCS Concepts

- Security and privacy → Information-theoretic techniques;
- Computing methodologies → Visual inspection.

## Keywords

Secret Image Distribution, Generative Image Steganography, Diffusion Model

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\*Corresponding author



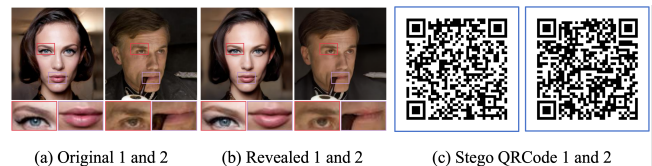
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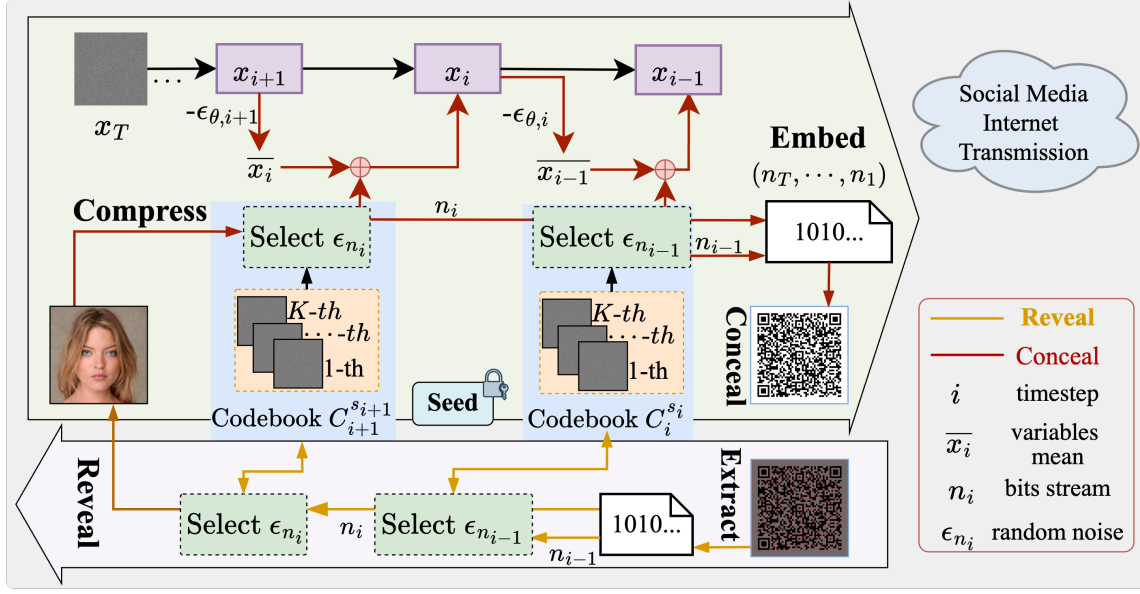


**Figure 1: Hiding high-resolution secret images in standard QR codes.** We take a pre-trained discrete diffusion model as a generation prior to achieve extreme compression, where the  $256 \times 256$  image in (a) is compressed into a bit stream of only 1200 bits. This ultra-compact payload can fit into a standard QR code, and even the smallest QR Version 5 is sufficient to accommodate it in (c). The extracted bitstream and high-fidelity restoration preserve the core semantic content and visual quality as shown in (b).

## 1 Introduction

In the era of ubiquitous connectivity, sensitive information such as faces, fingerprints, remote sensing, and pathological images is collected and transmitted extensively [42, 44]. The rapid expansion of digital visual content on the Internet and social media heightens the need for secure, covert communication that remains reliable through open, untrusted channels [37]. Protecting sensitive visual intelligence from unauthorized detection while ensuring accurate transmission has become a critical frontier [8, 34].

The existing secure transmission methodologies are divided into two distinct paradigms: encryption and steganography [1, 16]. While effective for textual data, multimedia has inherent characteristics of high redundancy and substantial data volume that render traditional encryption mechanisms suboptimal. Steganography hides secret data inside natural media by manipulating containers that often introduce subtle statistical anomalies [11, 21, 39]. These artifacts are imperceptible to humans, but present vulnerabilities that advanced steganalysis tools powered by modern deep learning can exploit. Natural images exhibit complex statistical regularities that advanced detectors can learn to characterize and identify even minor data-embedding deviations [2]. This inherent limitation forces an unsustainable trade-off: increasing payload raises detectable traces, while reducing capacity undermines practical utility for web-scale applications that must reliably transmit substantial visual information over potentially noisy channels [46]. The limitation stems from the reliance on natural media as containers [36].



**Figure 2: The framework of DiSCoQR, an end-to-end pipeline for secure image steganography through diffusion-driven semantic compression.** The secret image is first semantically compressed by a pre-trained diffusion model into a compact bitstream encoded with a discrete codebook. This bitstream is embedded in standard QR code patterns while preserving full scanner compatibility. The stego QR code takes its built-in error correction for robust transmission. The extracted bitstream directs a conditional reverse diffusion process, yielding high-fidelity, semantically aware images.

This vulnerability intensifies when compression, resizing, or format conversions disrupt embedding patterns. As AI advances, the gap between embedding sophistication and detection narrows. A fundamental redesign of steganographic containers moving beyond natural-media statistics is required [18].

Invisible but lasting generative image steganography (GIS) has become the core of AI-Generated Content (AIGC) security. Diffusion-based GIS takes secret-to-image reversible transforms to turn secrets into latent codes, boosting capacity and robustness. The recent adoption of diffusion models has opened new pathways by capitalizing on their noise resilience and zero-shot image translation capabilities [10, 28–30]. Diffusion models are probabilistic frameworks that learn a data distribution  $p(x)$  by gradually denoising Gaussian noise, effectively reversing a Markov chain process. It is trained to iteratively corrupt data by gradually adding Gaussian noise during the forward process. Denoising diffusion probabilistic models (DDPM) [10] transform a priori data distributions into random noise obeying the Gaussian distribution in the diffusion (forward) process. DDIM [29] accelerates the generation by an implicit probabilistic model with a deterministic forward process. Latent diffusion models (LDM) [28] operate in a low-dimensional latent space, reducing computational costs and fueling the rise of text-to-image tools, such as Stable Diffusion. They have demonstrated excellent performance in tasks such as image denoising, super-resolution reconstruction, and image deblurring, with their excellent ability to model data probability distributions [6, 11].

Existing diffusion-based GIS require inputs to conform to Gaussian distributions during diffusion sampling. Although distribution preservation [4, 38] can transform encrypted information into

Gaussian-like representations, the DDIM inversion [12, 33] for secret extraction remains vulnerable to distributional shifts and perturbations. This vulnerability is pronounced when stego-images undergo common transformations during transmission, where perturbations can amplify deviations during the reconstruction phase, leading to irreversible information loss. The inherent sensitivity of diffusion inversion to input variations creates a fundamental robustness bottleneck that limits real-world applicability.

To address these critical limitations, we propose a novel QR code-based framework (DiSCoQR) that rethinks the robustness in steganographic systems. Figure 1 shows that DiSCoQR successfully encodes and embeds a high-resolution natural image (size  $256 \times 256$ ) into a standard QR Code Version 5 (size  $62 \times 62$  modules). The stego QR code remains identifiable. By leveraging the structural resilience and error correction capabilities inherent in standardized QR codes, DiSCoQR circumvents the distributional constraints of pure diffusion-based methods, and takes QR codes as steganographic containers due to their ubiquity, standardized structure, and built-in error correction, ensuring reliable data retrieval even with significant distortion [3, 23, 32]. Since QR codes encode machine-readable data, any embedded payload—covert or not—follows expected statistical patterns rather than triggering anomalies. [5]. By grounding steganographic security in these normative properties, DiSCoQR decouples security from the statistical regularities of natural images, enabling a new paradigm for trustworthy information exchange, especially in web contexts where QR codes are a trusted, pervasive component of the digital ecosystem. Figure 2 shows the overall framework, which avoids hiding data in the statistical complexities of natural images and operates without model retraining.

This hybrid paradigm shifts the robustness burden from the diffusion process itself to the structurally robust QR code container, enabling reliable secret recovery in challenging transmission environments where conventional diffusion-based methods would inevitably fail. The main contributions of this work are threefold:

- We introduce a training-free framework for ultra-compact image compression. Departing from conventional approaches focused on pixel-level accuracy, our method leverages the inherent semantic understanding of pre-trained discrete diffusion models to distill high-resolution visual content into compact binary representations.
- We propose a robust steganographic framework that embeds high-capacity visual intelligence into standard QR codes while maintaining full functional compatibility.
- We conduct extensive experimental validation across multiple dimensions to rigorously establish the effectiveness of our proposed framework.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive review of related work in image steganography and generative models for reconstruction. Section 3 details our methodology, including the extreme compression framework, QR code embedding process, and reconstruction mechanism. Section 4 presents our experimental setup and results, including comparisons with state-of-the-art methods and ablation studies. Finally, Section 5 discusses the implications of our findings.

## 2 Related Work

**Cover-based Steganography.** Prior research predominantly employed the least significant bit or frequency domain to embed secrets into cover images [31, 45]. To prevent detectable traces after embedding, Inverted neural networks (INNs) model steganography with a single invertible network for both forward and inverse passes, avoiding performance loss from mismatches between separate encoding and decoding networks. HiNet [13] introduces a low-frequency loss to restrict hidden messages within high-frequency wavelet subbands. Wang et al. [35] redefines information hiding as a super-resolution problem. Lan et al. [19] integrate INN into an end-to-end robust steganographic system by embedding secrets into frequency domain coefficients. However, INNs introduce vulnerabilities to deep steganalysis due to their dependence on pre-existing covers and restricted capacity in image feature representation. Ren et al. [27] generate visually appealing QR code shares to avoid drawing attention during the distribution of images over public networks. While our method employs generative reconstruction based on compressed semantic representations. Robust Message Steganography (RMSteg) [40] takes tokenized image representations to imperceptibly embed QR code messages in host images, achieving strong robustness but limited capacity for concealing large visual intelligence compared to our extreme compression methods.

**Coverless Steganography.** Recent advances in diffusion-based steganography have demonstrated significant progress in coverless generation. Chen et al. [4] propose a rejection sampling strategy that encodes secret messages into latent vectors conforming to a Gaussian distribution. Yu et al. [39] introduce a reversible image-to-noise mapping framework for information embedding. CRoSS [43] utilizes open-source tools to improve controllability and diversity

of container images. Yang et al. [38] achieve theoretical guarantees of steganography by embedding information into Gaussian-conforming latent representations. StegaDDPM [25] utilizes the residual from the last non-noise step to embed messages. Instead of utilizing the noise at the last step, we leverage the indices of the noise codebook to embed information.

**Discrete Generative Modeling.** GIS conceals covert data by exploiting the inherent redundancy in pixel representations. Different from iterative sampling from a continuous Gaussian distribution, discrete generative modeling, such as vector-quantized latent tokens [9] and hierarchical modeling schemes [20] progressively refines each generated sample. Logits-based GIS [7] takes transformer tokens for secure data distribution, which requires shared autoregressive prefixes and is limited to small stego. In this paper, we take denoising diffusion codebook models (DDCM) [24] to obtain a discrete and finite noise representation of secret images, rather than traditional compression metrics based on actual file size. Unlike classic hide-in-pixel steganography, we decouple secrets from spatial pixels and model the hiding process as QR code encoding.

## 3 Proposed Method

### 3.1 Overview

**Problem Formulation.** Given a secret image  $\mathbf{I}_s \in \mathbb{R}^{H \times W \times 3}$ , the compressed bitstream as  $\mathbf{B} \in \{0, 1\}^L$ , the stego QR code as  $\mathbf{Q}_s \in \{0, 1\}^{M \times M}$ , and the reconstructed image as  $\hat{\mathbf{I}}_s \in \mathbb{R}^{H \times W \times 3}$ .  $\mathcal{C}$  stands for the compression function,  $\mathcal{E}$  for the embedding function,  $\mathcal{D}$  for the extraction function, and  $\mathcal{R}$  for the reconstruction function. This design ensures  $\mathbf{Q}_s$  matches standard QR codes in both visual appearance and functionality. It also meets the steganography:

$$\mathcal{D}(\mathcal{E}(\mathbf{B})) = \mathbf{B}', \quad \mathbf{B}' \triangleq \mathbf{B} \quad (1)$$

**Our Solution against Vulnerability.** Traditional compression methods optimize rate-distortion tradeoffs at the pixel level, whereas our approach leverages semantic understanding from pre-trained diffusion models to achieve superior perceptual quality at extreme compression ratios. The reconstructions reside on the same semantic manifold as the originals, ensuring their functional equivalence in downstream tasks. We integrate Reed-Solomon error correction for systematic protection against transmission distortions and employ discrete codebook-based semantic compression for reconstruction fidelity under quality degradation.

### 3.2 Diffusion-Driven Semantic Compression Steganography

**Diffusion Prior Compression.** We first take a discrete diffusion process to map continuous image space to compact binary representations while preserving semantic content. Given the outstanding performance of DDPM and DDIM in image generation, we apply them for steganography to generate high-quality stego images. For an input image  $\mathbf{x}_0 \sim p_{\text{data}}$ , the forward diffusion process follows a Markov chain. This chain progressively transforms the image into a noise distribution while capturing its essential semantic structure. The Markov chain's transition kernel iteratively adds controlled noise while preserving essential semantic information.

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (2)$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad (3)$$

where we construct a finite codebook  $C = \{\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \dots, \mathbf{z}^{(K)}\}$  where each  $\mathbf{z}^{(i)} \sim \mathcal{N}(0, \mathbf{I})$  and  $K = 2^k$ . Instead of storing continuous values, we project the diffusion process onto this codebook by selecting indices that optimally preserve semantic content.

**Bitstream Representation.** At each reverse diffusion step, we choose codebook entries to minimize semantic distortion. This selection takes deep feature distances to measure semantic dissimilarity, with  $\Phi$  representing a pre-trained feature extractor.

$$k_t = \arg \min_{j \in \{1, \dots, K\}} \mathcal{D}_{\text{semantic}}(\mathbf{x}_0, \hat{\mathbf{x}}_0(\mathbf{x}_t, \mathbf{z}^{(j)})) \quad (4)$$

$$\mathcal{D}_{\text{semantic}}(\mathbf{x}, \mathbf{y}) = \|\Phi(\mathbf{x}) - \Phi(\mathbf{y})\|_2^2 \quad (5)$$

where generates a compact bitstream  $\mathbf{B} = (k_1, k_2, \dots, k_T)$  with a total bit length of  $L = T \cdot \log_2 K$ , achieving an unprecedented compression ratio while retaining sufficient semantic information for high-quality reconstruction. The reverse process takes a modified diffusion equation, incorporating chosen codebook entries at each denoising step to maintain alignment with the original image content.

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t, k_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t) + \sigma_t \mathbf{z}^{(k_t)}, \tilde{\beta}_t \mathbf{I}\right) \quad (6)$$

where  $\mu_{\theta}(\mathbf{x}_t, t)$  is the predicted mean and  $\sigma_t$  controls the codebook injection strength. The compression model takes pre-trained diffusion models as fixed priors, requiring no extra training. It frames codebook selection as an optimization that minimizes semantic distortion under bitrate constraints, enabling extreme compression ratios unattainable by conventional methods.

### 3.3 Robust QR Code Steganography via Semantic Compression

**Bitstream Encoding.** Unlike conventional steganography, DiS-CoQR leverages the structural robustness of QR codes to create a reliable steganographic channel that withstands web-based transmission challenges, while remaining compatible with standard QR scanners. A QR code like Version 5 features a  $62 \times 62$  module grid and provides a data capacity  $C_{\text{max}} = 539$  bytes with low error correction. Our embedding function  $\mathcal{E} : \{0, 1\}^L \rightarrow \{0, 1\}^{62 \times 62}$  must adhere to structural constraints to ensure the resulting stego QR code remains valid. Let the compressed bitstream segment be represented as a message vector  $\mathbf{m} \in \mathbb{F}_{256}^k$ , where each element  $m_i \in \mathbb{F}_{256}$  corresponds to a byte of the bitstream. The RS encoding employs a generator matrix  $\mathbf{G} \in \mathbb{F}_{256}^{k \times n}$  to produce a codeword  $\mathbf{c} \in \mathbb{F}_{256}^n$ . The generator matrix  $\mathbf{G}$  is constructed from the generator polynomial  $g(x) = \prod_{i=0}^{n-k-1} (x - \alpha^i)$ , where  $\alpha$  is a primitive element of  $\mathbb{F}_{256}$  (typically  $\alpha = 2$  for QR code standardization).

The number of correctable errors  $t$  (per RS block) is determined by the code's redundancy  $(n - k)$ . For QR v5 with low error correction,  $n - k = 32$ , so  $t = 16$ . This means each RS block can correct up to 16 bytes of errors or 32 bytes of erasures, which is critical for preserving the steganographic payload during noisy transmission.

**QR Code Block Segmentation.** Rather than reusing the built-in error correction to repair scannable damage, we repurpose this capability to guarantee reliable transmission of hidden visual intelligence over noisy web channels. For payloads exceeding the RS block length  $n = 255$  bytes, the compressed bitstream  $\mathbf{B}$  is split into  $m$  consecutive message blocks  $\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_m$  (each of length  $k$ ). Each block is encoded into a codeword  $\mathbf{c}_i = \mathbf{m}_i \cdot \mathbf{G}$ , and the concatenated codewords  $[\mathbf{c}_1; \mathbf{c}_2; \dots; \mathbf{c}_m]$  are embedded into the QR code's data modules. This segmentation ensures compliance with QR code standards while maintaining error correction coverage across the entire payload. By balancing payload size and error correction overhead, we ensure steganographic content remains recoverable even under significant degradation that would disrupt conventional steganographic methods. DiS-CoQR transforms the QR code from a simple container into an active part of the steganographic process, where its native robustness directly enhances the security and reliability of covert communication. The stego-images are visually indistinguishable from legitimate QR codes and fully scannable by standard readers, decoupling steganographic security from the vulnerable statistical properties of natural images.

**Diffusion Guide Decoding.** Given the extracted bitstream  $\mathbf{B}'$ , we sample from the posterior distribution:

$$p(\mathbf{x}_0|\mathbf{B}') = \int p(\mathbf{x}_0|\mathbf{x}_{1:T})p(\mathbf{x}_{1:T}|\mathbf{B}')d\mathbf{x}_{1:T} \quad (7)$$

The reconstruction starts by initializing the reverse diffusion process with random noise. It then iteratively denoises the noise while injecting the codebook entries specified by the bitstream sequence. At each step, we compute the next estimate by combining the standard diffusion update with the codebook-specified direction, steering the reconstruction toward the semantic neighborhood of the original image. The reverse diffusion process with codebook guidance follows:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \tilde{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}^{(k'_t)} + \sqrt{\tilde{\beta}_t} \mathbf{w}_t \quad (8)$$

where  $\mathbf{w}_t \sim \mathcal{N}(0, \mathbf{I})$  and  $k'_t$  is the codebook index from the extracted bitstream. This conditional generation process enables the diffusion model to generate perceptually plausible details lost during compression while maintaining semantic fidelity to the original content.

**Plug-and-Play.** Our method exhibits a plug-and-play design that enables seamless integration into existing training-free conditional diffusion models. Within the standard diffusion sampling loop, we retrieve optimal noise vectors from predefined codebooks, followed by similarity computation to determine the best-matching codebook indices. This integration requires minimal code additions to the standard sampling pipeline while delivering significant improvements in output, including but not limited to DDPM, DDIM, and Stable Diffusion architectures. Unlike alternative methods that rely on complex gradient computations and adjustment mechanisms, our codebook-guided noise selection achieves more stable image generation with negligible computational overhead.

## 4 Experimental Evaluation

### 4.1 Experimental Setup

**Datasets.** We conduct a comprehensive evaluation across four benchmark datasets that address the core challenges of modern web applications, including social media content sharing, digital documentation, and cross-platform visual communication. The CelebA-HQ dataset [14] provides 30,000 high-resolution celebrity portraits at  $256 \times 256$  resolution, serving as the primary benchmark for human facial analysis. The FFHQ dataset [15] contains 70,000 high-quality facial images with greater diversity in age, ethnicity, and background, offering enhanced evaluation of robustness. The Flickr30K dataset comprises 31,000 natural scene images with rich contextual elements. Kodak images provide a standardized benchmark for evaluating steganographic methods in latent space, consisting of 24 high-quality images that are center-cropped to  $512 \times 512$  for compatibility with Stable Diffusion [17].

**Evaluation Metrics.** For *perceptual quality*, we employ Fréchet Inception Distance (FID) to measure distribution-level similarity between generated and real images, NIQE [22], and Inception Score (IS) to evaluate both visual quality and diversity of generated stego-images. For *semantic consistency*, we utilize CLIP-Score [26] to quantify the semantic alignment between original and reconstructed content. *steganographic security* is evaluated using detection accuracy rates against steganalysis frameworks including SRNet [2] and SiaStegNet [41]. The performance metrics adopted in our experiments include true positive rate (TPR) and False Positive Rate (FPR) to measure extraction performance.

**Baseline Methods.** We conduct extensive comparisons with representative steganographic approaches spanning different methodologies. Traditional approaches include 4-bit LSB embedding as the conventional baseline. Deep learning-based methods comprise HiDDeN [48] with its recommended architecture featuring 64-channel bottlenecks, SteganoGAN [25] using published configurations, and Baluja’s method [1] for cover-based hiding. Diffusion-based approaches include the recently proposed StegaDDPM [25] and CRoSS [43], representing the state-of-the-art in generative steganography. All methods are evaluated by their official implementations with optimal hyperparameters for fair comparison. We additionally compare with several compression and reconstruction methods: JPEG-1 employing traditional JPEG compression with quality factor of 1, Walsh-Hadamard Transform-based Compressed Sensing (WHT-CS), CS image reconstruction method using bilinear interpolation, and Low-Rank Guided Diffusion (LRGD) for robust image transmission in semantic communication [47].

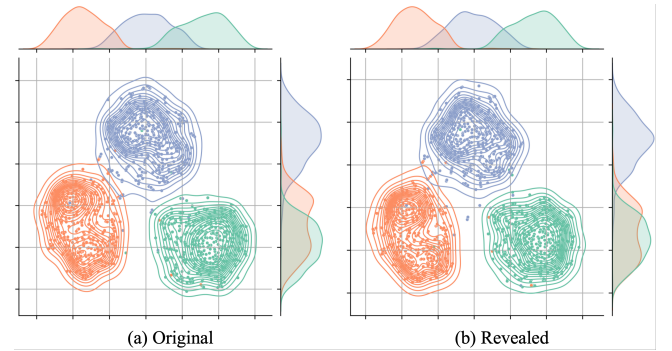
**Implementation Details.** Our framework builds upon pre-trained diffusion models as generative priors without requiring additional training. We adopt the default setting of DDIMPipeline. To enable invertible image translation, we configure the guidance scale parameter in Stable Diffusion to 1. All experiments take a base seed of 42 for reproducibility and are conducted on an Ubuntu workstation with an Nvidia RTX4090 24GB GPU.

### 4.2 Performance Analysis

Table 1 shows our method establishes state-of-the-art performance by fundamentally rethinking the steganographic paradigm through QR code integration with discrete diffusion models. Traditional

steganography methods operate within a constrained design space, inevitably creating trade-offs. CRoSS exemplifies this limitation through its reliance on deterministic DDIM inversion, which creates an irreversible information bottleneck that severely compromises reconstruction fidelity for semantically rich content. Similarly, HiNet and other inversion-based approaches suffer from inherent information loss that forces an artificial choice between reconstruction accuracy and semantic preservation. Diffusion-based methods like StegaDDPM demonstrate competitive generation quality but exhibit significant semantic drift, indicating inadequate modeling of semantic relationships during the embedding process. The diffusion architecture, while powerful for generation, lacks explicit mechanisms for semantic preservation when adapted for steganography. GAN-based approaches, including SteganoGAN, show a critical divergence between perceptual quality and semantic consistency. This exposes their inherent bias toward texture synthesis over genuine semantic understanding, limiting their effectiveness for content-aware steganography. Deep-based methods such as Baluja and HiDDeN face architectural constraints in adapting to complex visual content. Their fixed embedding strategies cannot adequately account for contextual variations, resulting in suboptimal performance across diverse datasets.

DiSCoQR surpasses traditional trade-offs by reframing steganography as guided exploration of the semantic manifold rather than just visual perturbation. The QR structure reinforces a robust information container, while the diffusion process preserves semantic coherence. This dual benefit is evident in latent-space evaluations on the Kodak dataset, where our method remains strong despite the reconstruction challenges of VAE encoding.



**Figure 3: Joint distribution of original samples and revealed samples for three expression attributes. The distributions of the revealed samples by our method and the original samples are similar.**

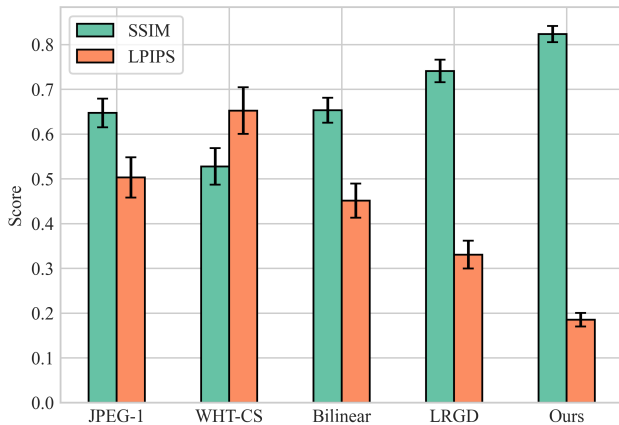
**Feature-Space Fidelity.** We analyze the joint distribution of original samples and their reconstructions in Figure 3. Although the reconstructions are not pixel-perfect, their feature-space distributions align closely and achieve functional equivalence, preserving the essential semantic characteristics of the originals in three deep expression attribute features. This distributional similarity ensures reconstructed images remain functionally equivalent for downstream tasks such as classification, detection, and semantic analysis. The perceptual quality and feature-space consistency achieved by

**Table 1: Quantitative comparison of steganographic performance across pixel space and latent space.**

| Method     | Pixel Space |       |      |       |       |      |        |       |      | Latent Space |       |      |
|------------|-------------|-------|------|-------|-------|------|--------|-------|------|--------------|-------|------|
|            | Celeba      |       |      | FFHQ  |       |      | Flickr |       |      | Kodak        |       |      |
|            | FID↓        | CLIP↑ | IS↑  | FID↓  | CLIP↑ | IS↑  | FID↓   | CLIP↑ | IS↑  | FID↓         | CLIP↑ | IS↑  |
| Ours       | 8.45        | 0.89  | 3.45 | 9.12  | 0.89  | 3.38 | 7.89   | 0.90  | 3.52 | 15.23        | 0.85  | 3.18 |
| HiNet      | 12.34       | 0.84  | 3.12 | 13.56 | 0.83  | 3.05 | 11.87  | 0.85  | 3.18 | 18.45        | 0.81  | 2.95 |
| StegaDDPM  | 15.78       | 0.81  | 2.95 | 16.45 | 0.80  | 2.87 | 14.92  | 0.82  | 3.01 | 21.67        | 0.79  | 2.78 |
| CRoSS      | 45.23       | 0.62  | 2.15 | 47.12 | 0.60  | 2.08 | 43.87  | 0.64  | 2.21 | 52.34        | 0.60  | 1.89 |
| Baluja     | 28.67       | 0.73  | 2.67 | 29.45 | 0.72  | 2.59 | 27.34  | 0.75  | 2.72 | 35.12        | 0.69  | 2.45 |
| HiDDeN     | 32.45       | 0.69  | 2.45 | 33.78 | 0.67  | 2.38 | 31.23  | 0.70  | 2.51 | 38.89        | 0.65  | 2.28 |
| SteganoGAN | 38.12       | 0.65  | 2.28 | 39.45 | 0.63  | 2.21 | 36.78  | 0.66  | 2.35 | 43.23        | 0.61  | 2.12 |
| 4-bit LSB  | 52.34       | 0.52  | 1.87 | 54.12 | 0.50  | 1.79 | 50.45  | 0.55  | 1.92 | 58.67        | 0.52  | 1.65 |

our approach demonstrate its practical utility for applications where semantic fidelity matters more than exact pixel-level reconstruction. The reconstructed data is not an approximation, but a semantically consistent representation that is inherently more resilient. Perfect pixel recall is not only infeasible but unnecessary. True fidelity lies in the preservation of semantic topology.

**Semantic Communication Analysis.** Figure 4 shows varying semantic compression performance. Traditional methods struggle under extreme compression. JPEG-1 keeps structure but loses semantics due to pixel-focused encoding. WHT-CS performs even worse at preserving semantic meaning. Bilinear offers basic recovery but misses key semantic details, yielding visually plausible yet semantically weak results, underscoring the limits of purely geometric methods for semantic tasks. The gap to existing methods highlights the need to align compression with semantic preservation, not just signal accuracy. LRGD adds structural priors, improving semantic preservation but still failing to retain fine-grained semantics across content. Our approach takes discrete diffusion with codebooks, surpassing others in perceptual quality and semantic fidelity.

**Figure 4: comparative analysis of reconstruction quality and semantic preservation across compression methods.**

### 4.3 Security Analysis

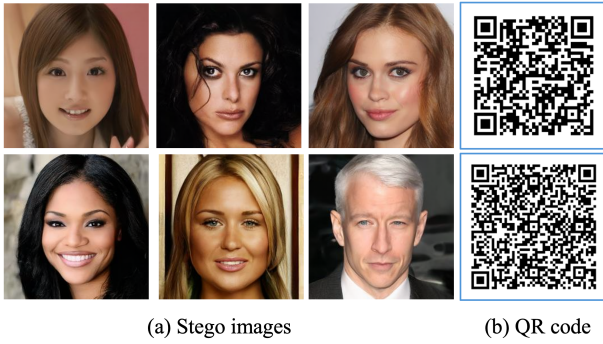
**Steganalysis Detection.** We verify the undetectability of our steganographic QR codes against both structural analysis and statistical steganalysis. Figure 7 shows DiSCoQR provides inherent immunity against modern steganographic detection. Traditional steganography becomes increasingly detectable under advanced analyses like SRNet, often approaching near-perfect identification. In contrast, our method sustains detection at the random-guessing level throughout training. By exploiting QR codes as the container, our framework redefines steganographic security. Although QR codes have limited native capacity, our extreme compression model distills natural images into compact binary streams of a few hundred bits, enabling the transmission of stego-content. The resulting stego-images fully conform to QR code specifications, remaining visually and structurally indistinguishable from standard codes. Quantitative results in Table 2 further confirm the consistent un-

**Table 2: Comparison of methods under steganalysis detection (SRNet[2], SiaStegNet[41]) and NIQE scores.**

| Method   | NIQE↓       | SRNet↓ | SiaStegNet↓ |
|----------|-------------|--------|-------------|
| 4bit-LSB | 6.11 ± 1.74 | 1.00   | 0.99        |
| Baluja   | 3.48 ± 0.93 | 0.99   | 0.95        |
| WengNet  | 3.85 ± 1.20 | 0.75   | 0.76        |
| HiNet    | 5.57 ± 1.99 | 0.56   | 0.55        |
| PUSNet   | 3.55 ± 1.01 | 0.58   | 0.82        |
| HiDDeN   | 3.56 ± 0.89 | 0.76   | 0.77        |
| CRoSS    | 3.04 ± 0.92 | 0.55   | 0.54        |
| Ours     | 4.12 ± 2.85 | 0.50   | 0.50        |

detectability across multiple advanced steganalysis platforms. The security superiority stems from the fundamental paradigm shift our method represents. By exploiting the standardized QR code framework, we separate steganographic security from the natural-image statistics disrupted by conventional methods. This removes the main attack vector used by modern deep-learning steganalyzers that rely on detecting statistical anomalies in image features.

**Seed Analysis.** The embedded visual information remains unrecoverable without authorized shared seeds. Figure 5-(a) shows distinct portrait variations including individuals with dark curly hair, light



**Figure 5: Cryptographic steganography enables identical QR codes to generate diverse facial reconstructions through seed variation while maintaining full machine readability.**

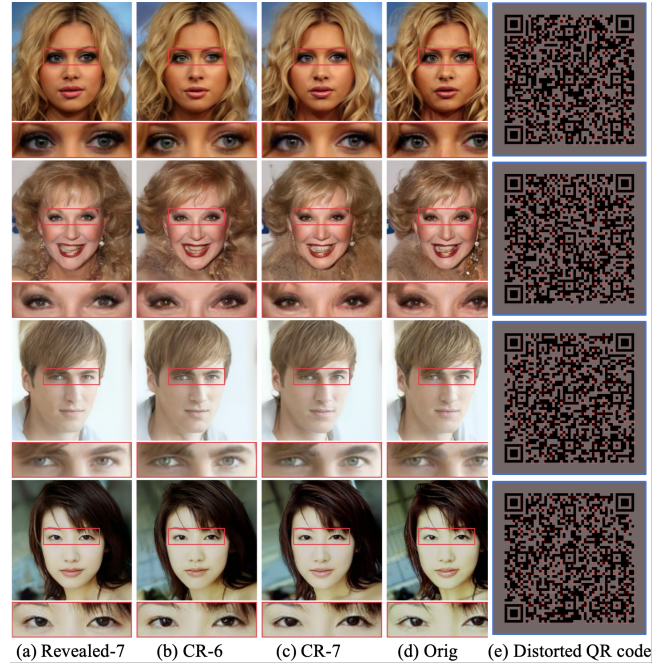
brown wavy hair, blonde hair, and a white-haired gentleman in a formal suit, all derived from identical QR code structures in Figure 5-(b) while maintaining identical machine-readable properties. DiSCoQR achieves perfect steganographic security. Different cryptographic seeds reconstruct distinct images from the same encoded message. This leaks no detectable patterns or statistical artifacts.

#### 4.4 Robustness Analysis

Figure 6 shows the robustness of our secret image revealing method against JPEG compression attacks. The experiment tests performance under severe compression with a quality factor of 20, a substantial distortion. For the four original secret images, the results indicate that our method reveals recognizable secret images from the heavily compressed QR codes, despite artifacts from aggressive compression seen in column Figure 6-(e). Comparisons with revealed images taking cookbook sizes of  $2^6$  and  $2^7$  in columns (b) and (c) highlight the optimization and efficiency of our method. This robustness holds across various image types, including natural skin tones and hair colors that are challenging for compression algorithms.

We further evaluate robustness quantitatively against nine common perturbations, including compression, cropping, noise, filtering, resizing, and brightness adjustment. Table 3 reports PSNR under each attack as well as the average drop relative to clean conditions. Our QR code-based method demonstrates exceptional resilience under JPEG compression, brightness adjustment, and random dropout attacks, achieving the highest PSNR scores among all compared methods in these challenging scenarios.

The inherent error correction capability of QR codes plays a pivotal role in this robust performance. The encoding scheme provides systematic protection against various distortions, enabling reliable data recovery even under severe degradation. This is evident in compression and brightness attacks, where our method maintains nearly identical performance to clean conditions, demonstrating the ability to withstand quality degradation that disrupts conventional steganography. Our method shows moderate performance under spatial transformations like resizing and cropping—areas where QR code’s structural patterns face inherent challenges. This



**Figure 6: Revealed imaged and decoded QR Codes under JPEG distortion. The distorted QR codes are fully scannable and recognized. (a) shows the secret image revealed from the lossy QR code in (e). Figures (b) and (c) are the revealed images with candidate cookbook sizes of  $2^6$  and  $2^7$ , respectively. Figure (d) is the original secret image, and the red denotes the distortion. The attack applied in Figure (e) is JPEG compression with quality= 20. All the sample steps are 200. CR denotes revealed images from a clean QR code.**

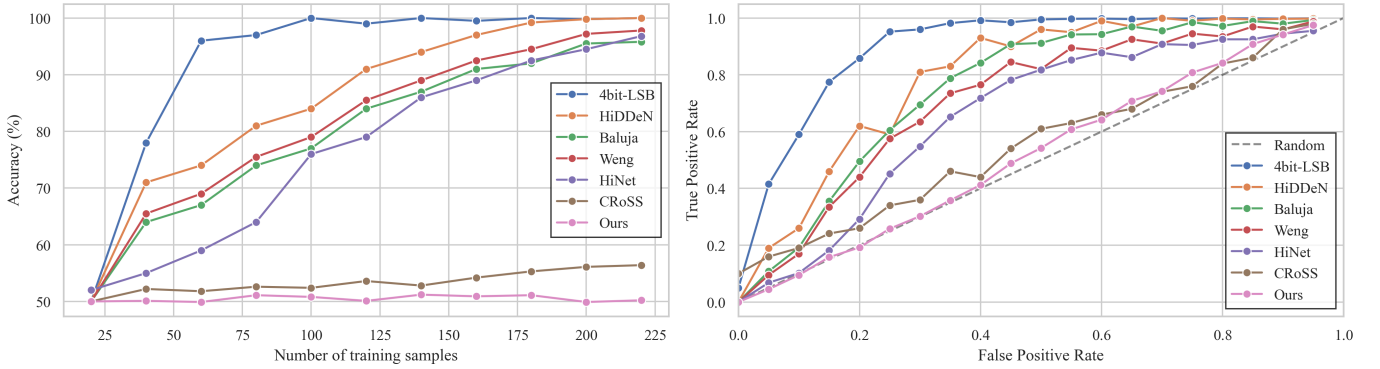
balanced performance profile confirms that our framework successfully leverages QR code’s error correction strengths while maintaining acceptable resilience.

#### 4.5 Ablation Study

Table 4 analyzes how diffusion sampling steps and codebook sizes affect steganographic performance. The reconstruction casts image recovery as a conditional generation task guided by the semantic content of the extracted bitstream. Using the QR code’s compressed representation, we exploit the generative prior of pre-trained diffusion models to reconstruct a high-fidelity approximation of the original secret image. Increasing diffusion steps consistently improves perceptual quality and semantic preservation, as shown by gains across all evaluation metrics. Expanding the codebook size also enhances reconstruction fidelity, since larger discrete spaces enable more accurate semantic encoding and recovery. Performance improves monotonically with more resources and capacity, though returns diminish at higher configurations. QR code version requirements scale predictably with total information capacity, keeping our approach practical across settings. These findings show that our framework offers flexible trade-offs between computational cost and quality while preserving the advantages of QR code-based

**Table 3: Quality comparison under nine attacks. Metric: PSNR (dB). Avg. denotes the average PSNR over the nine attacks, and ↓ (%) denotes the relative drop of the average PSNR compared with the clean case.**

| Methods  | clean | JPEG  | RandCrop | RandDrop | GauBlur | MedianFilter | GauNoise | SaltPepper | Resize | Brightness | Avg.  | ↓ (%) |
|----------|-------|-------|----------|----------|---------|--------------|----------|------------|--------|------------|-------|-------|
| 4bit-LSB | 29.16 | 29.16 | 9.17     | 19.26    | 15.79   | 27.80        | 7.83     | 19.04      | 10.25  | 7.59       | 16.65 | 42.9% |
| Baluja   | 34.24 | 16.52 | 10.57    | 22.64    | 17.35   | 20.96        | 11.58    | 8.27       | 13.25  | 16.35      | 15.94 | 53.5% |
| HiDDeN   | 28.19 | 19.49 | 11.01    | 22.94    | 19.69   | 15.87        | 19.58    | 13.74      | 15.82  | 18.55      | 17.74 | 37.1% |
| WengNet  | 39.75 | 17.76 | 12.56    | 21.97    | 16.23   | 19.50        | 12.95    | 8.07       | 12.70  | 14.03      | 15.75 | 60.4% |
| HiNet    | 42.98 | 7.03  | 10.38    | 14.49    | 15.13   | 14.90        | 12.91    | 13.50      | 12.00  | 11.00      | 12.15 | 71.7% |
| CRoSS    | 23.79 | 21.74 | 12.00    | 15.09    | 15.32   | 15.11        | 21.89    | 22.00      | 23.00  | 22.00      | 18.68 | 21.5% |
| Ours     | 28.29 | 27.90 | 13.53    | 24.10    | 13.12   | 13.10        | 13.11    | 13.24      | 13.15  | 26.50      | 17.64 | 37.6% |

**Figure 7: Comparative evaluation of anti-steganalysis performance. The left shows the detection accuracy evolution during SRNet training and the right shows the ROC curves.****Table 4: Ablation study on diffusion steps and codebook sizes. Performance is evaluated using FID ↓ and CLIP-Score ↑ metrics, with corresponding QR code versions indicated.**

| Diffusion Steps | Codebook Size  | Performance Metrics |              | QR Version |
|-----------------|----------------|---------------------|--------------|------------|
|                 |                | FID ↓               | CLIP-Score ↑ |            |
| 100             | 2 <sup>5</sup> | 15.23               | 0.821        | 3          |
|                 | 2 <sup>6</sup> | 12.45               | 0.845        |            |
|                 | 2 <sup>7</sup> | 10.67               | 0.862        |            |
|                 | 2 <sup>8</sup> | 9.34                | 0.878        |            |
| 200             | 2 <sup>5</sup> | 13.56               | 0.835        | 4          |
|                 | 2 <sup>6</sup> | 10.89               | 0.856        |            |
|                 | 2 <sup>7</sup> | 9.12                | 0.871        |            |
|                 | 2 <sup>8</sup> | 8.45                | 0.892        |            |
| 300             | 2 <sup>5</sup> | 12.34               | 0.842        | 5          |
|                 | 2 <sup>6</sup> | 9.78                | 0.864        |            |
|                 | 2 <sup>7</sup> | 8.23                | 0.883        |            |
|                 | 2 <sup>8</sup> | 7.89                | 0.901        |            |
| 500             | 2 <sup>5</sup> | 11.89               | 0.848        | 7          |
|                 | 2 <sup>6</sup> | 9.45                | 0.869        |            |
|                 | 2 <sup>7</sup> | 8.01                | 0.888        |            |
|                 | 2 <sup>8</sup> | 7.56                | 0.912        |            |

steganography. By harnessing a generative prior, our work effectively compresses the semantic entropy of an image while discarding perceptual redundancy.

## 5 Conclusions

In this paper, we propose a new approach using the structural robustness of standardized QR codes. The framework demonstrates that visual intelligence can be effectively encoded as semantic tokens rather than pixel-level representations, achieving unprecedented compression ratios while preserving reconstruction quality. The machine-readable nature of QR codes provides built-in error correction that naturally mitigates transmission degradation, while their predictable statistical distribution reduces detectability through conventional steganalysis. Existing steganography is limited by its reliance on vulnerable natural media. Our method decouples steganographic security from the statistical properties of natural images, neutralizing conventional steganalysis attacks. This shifts the security focus from natural image statistics to normative patterns engineered for machine interpretation. Comprehensive experiments show that our method produces a semantically consistent and inherently robust representation, yielding superior perceptual quality, semantic preservation, and undetectability.

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